

The background features abstract geometric designs in the corners. The top-left and bottom-left corners have overlapping light blue rectangles and diagonal lines. The top-right and bottom-right corners feature a grid of small light blue circles. The central text is white and stands out against the light blue background.

BLUESTOCK MUTUAL FUND ANALYTICS

Comprehensive Data Analytics,
Performance & Risk Analysis

By AMAL A E

SUMMARY

The Bluestock Mutual Fund Analytics project focuses on analysing mutual fund data to understand fund performance, investor behaviour, SIP trends, and portfolio risk. The project combines data cleaning, exploratory data analysis, financial analytics, and interactive Power BI dashboards to provide useful insights for investors and fund analysis. The analysis was performed using mutual fund master data, NAV history, fund performance data, SIP inflows, investor transactions, and portfolio holdings. Python and Pandas were used for data cleaning, transformation, exploratory analysis, and advanced analytics. Power BI was used to develop an interactive dashboard consisting of multiple analytical pages. The project evaluated important performance and risk metrics such as returns, volatility, Sharpe ratio, maximum drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and sector concentration using the Herfindahl-Hirschman Index (HHI). Investor analysis included cohort analysis and SIP continuation analysis to identify investment patterns and investors who may be at risk of discontinuing their SIPs. A simple fund recommendation model was also developed to identify the top-performing funds by Sharpe ratio according to the investor's risk category. The Power BI dashboard provides interactive visualizations for industry trends, fund performance, investor analytics, SIP inflows, market trends, and fund-level NAV analysis. Overall, the project demonstrates how financial data analytics can be used to evaluate mutual fund performance, identify investment and concentration risks, understand investor behaviour, and support data-driven investment analysis.

DATA SOURCES & DATASET DESCRIPTION

The Bluestock Mutual Fund Analytics project uses multiple datasets covering mutual fund information, historical NAV data, fund performance, SIP activity, investor transactions, market benchmarks and portfolio holdings. The datasets were stored in a SQLite database and cleaned CSV files for analysis.

DATA SOURCES

Dataset	Description	Main Purpose
Fund Master	Contains basic information about mutual fund schemes, fund houses, categories, plans, risk categories, expense ratios and minimum investment amounts.	Fund classification and fund-level analysis
NAV History	Contains historical Net Asset Value (NAV) observations for mutual fund schemes.	NAV trends and fund performance
AUM by Fund House	Contains Assets Under Management information for different fund houses.	Industry AUM and fund-house comparison
Monthly SIP Inflows	Contains monthly SIP inflow information	SIP trend and monthly inflow analysis
Category Inflows	Contains inflow information by mutual fund category	Category-wise inflow analysis
Industry Folio Count	Contains industry-level folio/account information	Industry investor growth analysis
Scheme Performance	Contains fund returns and risk-adjusted performance metrics	Fund performance and risk analysis
Investor Transactions	Contains investor-level SIP, Lumpsum and Redemption transactions.	Investor behaviour and cohort analysis
Portfolio Holdings	Contains individual fund holdings, sectors, weights and market values	Sector concentration and HHI analysis
Benchmark Indices	Contains benchmark market index data used for comparison with fund performance	Fund vs benchmark and market trend analysis

IMPORTANT FILEDS

Some of the important fields used during the analysis include:

FUND MASTER amfi_code scheme_name fund_house category sub_category plan risk_category expense_ratio_pct min_sip_amount benchmark	NAV HISTORY amfi_code date nav INVESTOR TRANSACTIONS investor_id transaction_date transaction_type amount_inr amfi_code city age_group gender	SCHEME PERFORMANCE amfi_code scheme_name return_lyr_pct return_3yr_pct return_5yr_pct alpha beta sharpe_ratio sortino_ratio std_dev_ann_pctm ax_drawdown_pct aum_crore risk_grade	THE TRANSACTION TYPE INCLUDES: SIP Lumpsum Redemption PORTFOLIO HOLDINGS amfi_code stock_symbol stock_name sector weight_pct market_value_cr current_price_inr portfolio_date
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DATABASE

The project data was also maintained in a SQLite database
“bluestock_mf.db”

DATA COVERAGE

The analysis covers mutual fund and investor activity across the project period, with historical data used to analyse trends from approximately 2022 to 2025. The data was used to study fund performance, industry AUM, SIP inflows, investor transactions, and portfolio concentration.

DATA PREPARATION

Before analysis, the datasets were inspected for missing values, incorrect data types, duplicate records and inconsistent values. Dates were converted into appropriate datetime formats, numerical columns were converted to numeric types, and relevant datasets were merged using common identifiers such as `amfi\_code`.

The cleaned datasets were then used for exploratory data analysis, advanced analytics, and Power BI dashboard development.



DATA ETL & CLEANING PROCESS

The project datasets were loaded into Python using Pandas and organized in the SQLite database `bluestock\_mf.db`.

THE MAIN ETL STEPS INCLUDED:

- Inspected all datasets for structure, missing values, duplicates and data types.
- Converted date columns into datetime format and numerical columns into appropriate numeric types.
- Cleaned and standardized the datasets for analysis. - Connected related datasets using common fields such as `amfi\_code`.
- Created analytical fields such as daily returns, investor cohort year, age groups and SIP transaction gaps.
- Prepared datasets for advanced analysis including VaR, CVaR, rolling Sharpe ratio and sector HHI.
- Validated the cleaned data before loading it into Power BI.

The processed datasets were then used for EDA, advanced analytics and the development of the four-page Power BI dashboard.

EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed to understand the structure, trends and patterns in the mutual fund data.

KEY AREAS ANALYSED INCLUDED:

- Industry AUM and folio growth over time.
- SIP inflow trends across months and categories.
- Fund returns, volatility and risk measures.
- NAV movement of selected mutual funds.
- Investor transaction patterns including SIP, Lumpsum and Redemption.
- Fund-house and category-wise performance.
- Portfolio sector allocation and concentration.

The EDA helped identify important trends and relationships that were further investigated through advanced analytics and Power BI dashboards.

FUND PERFORMANCE ANALYSIS

Mutual fund performance was analysed using historical returns, NAV trends and risk-adjusted performance indicators.

THE ANALYSIS INCLUDED:

- 1-year, 3-year and 5-year returns.
- Sharpe and Sortino ratios to evaluate risk-adjusted performance.
- Annualized standard deviation to measure volatility.
- Maximum drawdown to identify downside risk.
- Fund performance compared with relevant benchmark indices.
- A sortable fund scorecard and NAV trend visualization were developed in Power BI.

These measures helped compare funds based on both returns and the level of risk taken to generate those returns.

ADVANCED ANALYTICS & RISK ANALYSIS

Advanced analytics were performed to evaluate fund risk, investor behaviour and portfolio concentration.

KEY ANALYSES INCLUDED:

- VaR & CVaR: Measured potential downside risk using historical fund returns.
- Rolling Sharpe Ratio: Calculated 90-day rolling Sharpe ratios to monitor changes in risk-adjusted performance.
- Investor Cohort Analysis: Compared 2024 and 2025 investor cohorts based on SIP activity, total investment and fund preference.
- SIP Continuation Analysis: Identified investors with 6+ SIP transactions and flagged those with an average transaction gap above 35 days as at-risk.
- Fund Recommendation: Recommended the top 3 funds by Sharpe ratio based on investor risk appetite.
- Sector HHI: Measured sector concentration for equity funds using portfolio sector weights.

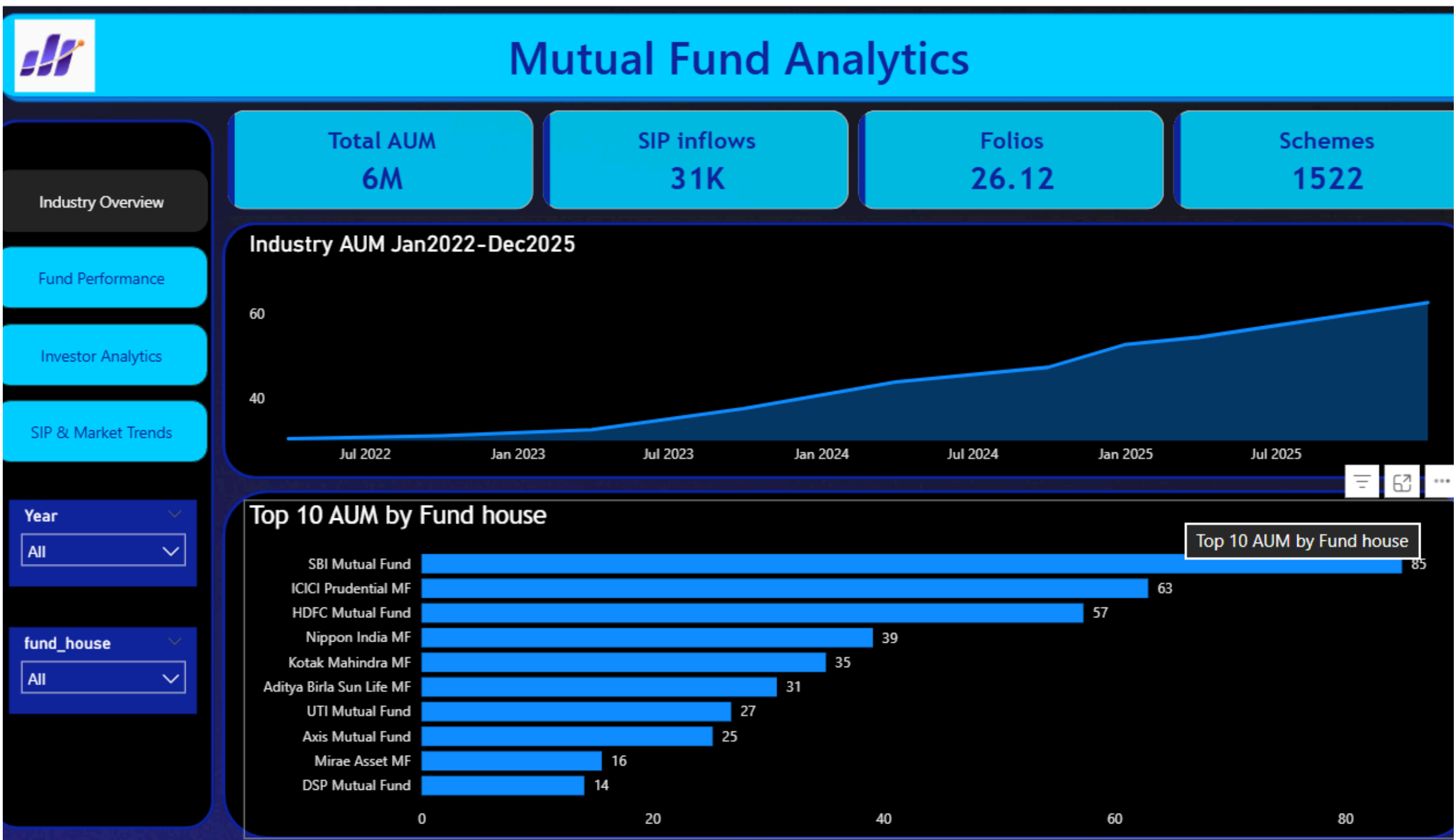
These analyses provided deeper insights into investment risk, investor behaviour, fund selection and portfolio diversification.

POWER BI DASHBOARD

An interactive Power BI dashboard was developed to present the key findings in a clear and user-friendly format.

PAGE 1 — INDUSTRY OVERVIEW

This page provides an overall view of the mutual fund industry.



- Key Visuals:
- kPI Cards: Total AUM, SIP Inflows, Folios and Number of Schemes.
 - Line Chart: Shows the trend in industry AUM from 2022 to 2025.
 - Bar Chart: Displays the top 10 fund houses based on AUM.
 - Slicers: Year and Fund house

Purpose:

To understand the overall size, growth and major participants in the mutual fund industry.

PAGE 2 — FUND PERFORMANCE

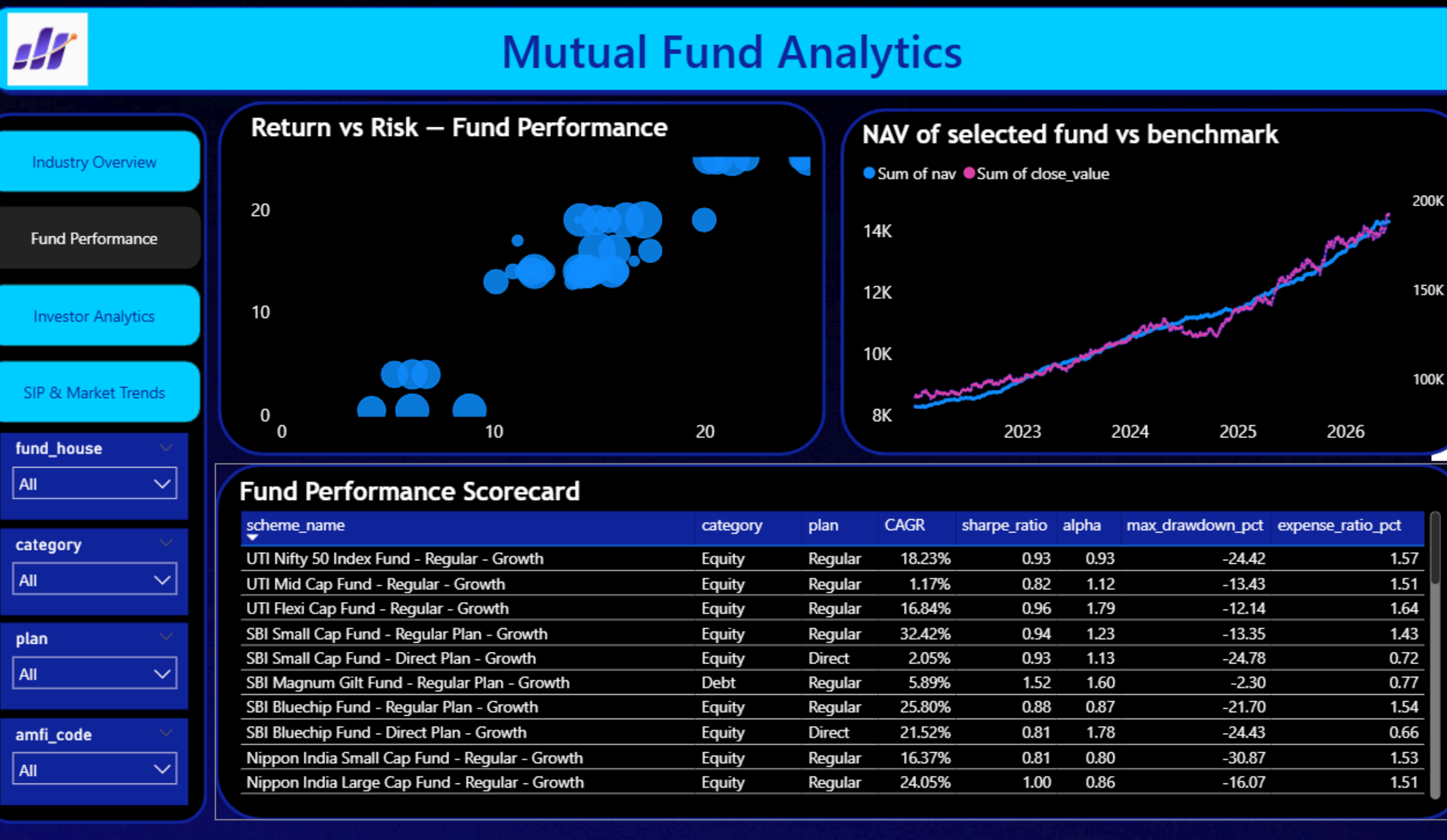
This page focuses on comparing individual mutual fund performance and risk.

Key Visuals:

- Scatter Plot: Compares fund returns with risk/standard deviation.
- NAV Line Chart: Shows the selected fund's NAV trend and benchmark comparison.
- Fund Scorecard: Provides a sortable view of fund performance metrics.
- Slicers: Fund House, Category ,Plan and amfi code

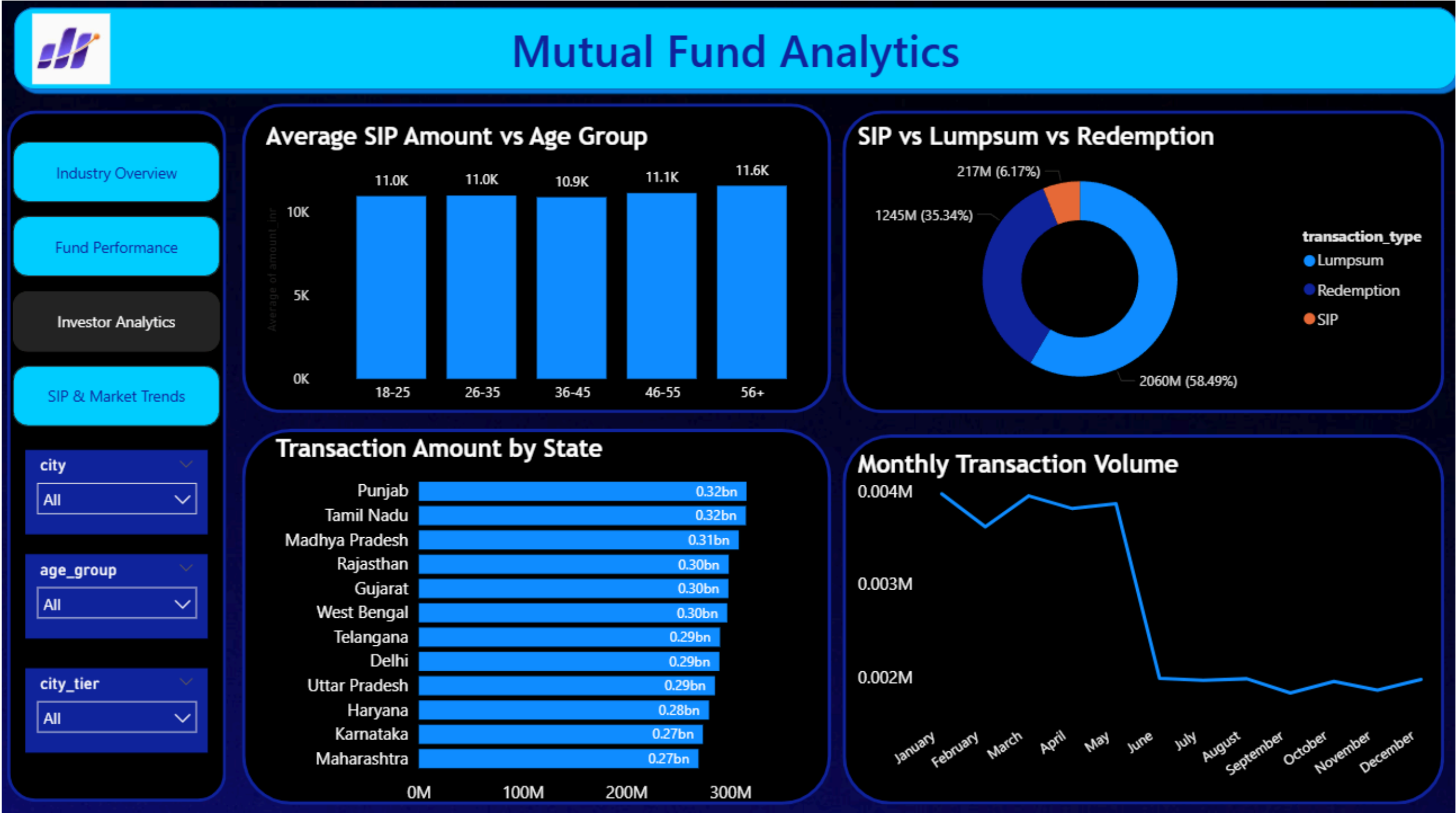
Purpose:

To help compare funds based on returns, risk, AUM and risk-adjusted performance.



PAGE 3 — INVESTOR ANALYTICS

This page analyses investor transaction behaviour.



Key Visuals:

- Bar Chart: Compares age groups based on average SIP amount.
- Donut Chart: Shows the proportion of SIP, Lumpsum and Redemption transactions.
- Bar Chart: Shows transaction amounts by state.
- Line Chart: Shows monthly transaction volume.
- Slicers: State, Age Group and City Tier

Purpose:

To understand investor demographics, transaction patterns and SIP behaviour.

PAGE 4 — SIP & MARKET TRENDS

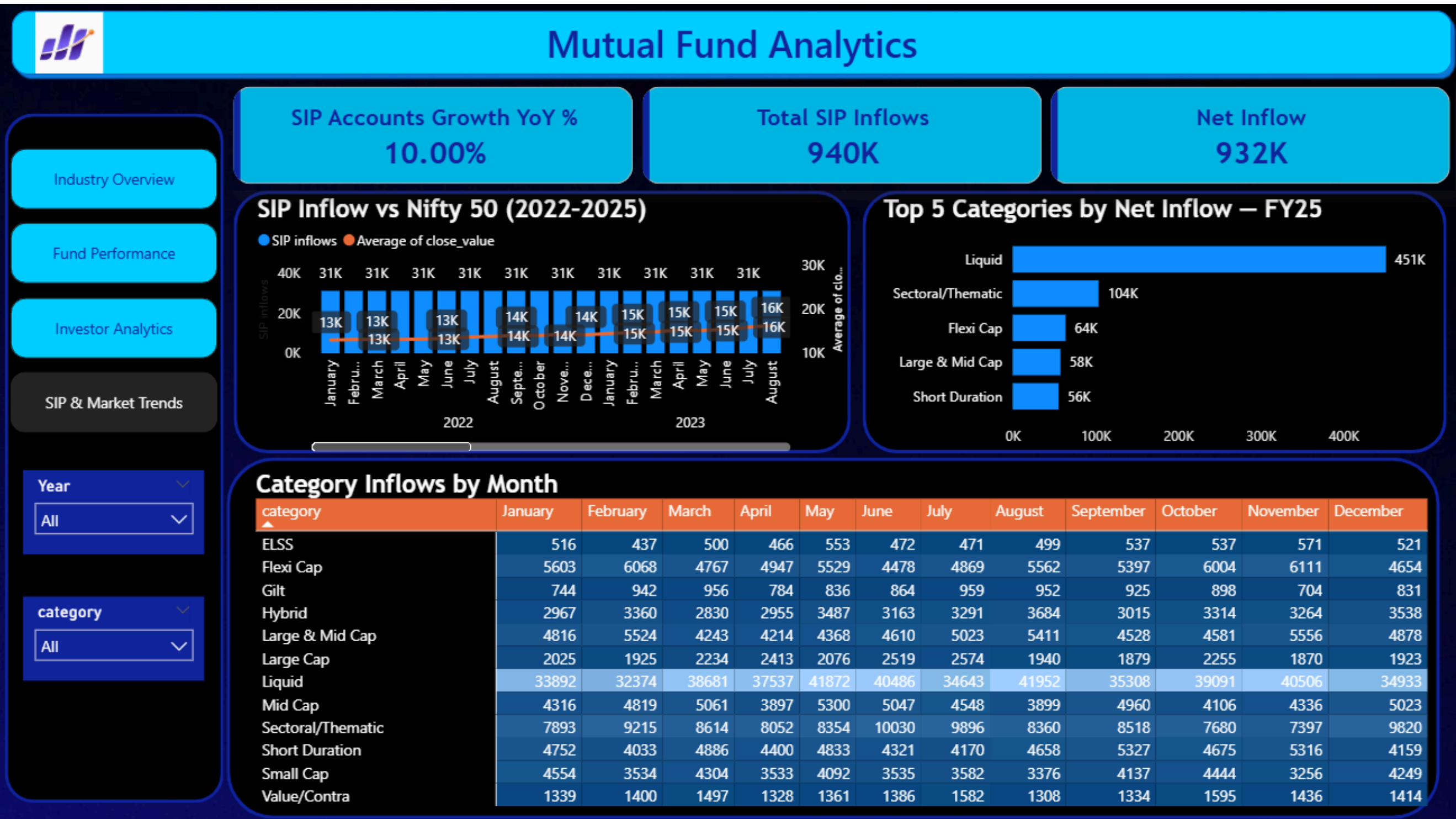
This page focuses on SIP activity and its relationship with market movements.

Key Visuals:

- Dual-Axis Chart: SIP inflows as bars and Nifty 50 as a line from 2022–2025.
- Bar Chart: Displays the top 5 categories by net inflow for FY25.
- Heat Map: Shows category-wise SIP/inflow patterns across months.
- KPI Card: Shows SIP account growth YoY, Total SIP Inflows and Net inflow
- Slicers: Year and Category

Purpose:

To identify SIP growth patterns, category preferences and the relationship between SIP activity and market trends.



KEY FINDINGS & RECOMMENDATIONS

KEY FINDINGS

- **Industry Growth:** AUM, SIP inflows and folio counts showed important changes over the analysis period, indicating the growth and changing participation in the mutual fund industry. **Rolling Sharpe Ratio:** Calculated 90-day rolling Sharpe ratios to monitor changes in risk-adjusted performance.
- **Fund Performance:** Funds differed significantly in returns and risk. Sharpe and Sortino ratios helped identify funds with stronger risk-adjusted performance.
- **Investor Behaviour:** SIP and Lumpsum transactions formed a major part of investor activity, while cohort analysis showed differences between investors who started investing in 2024 and 2025.
- **SIP Continuation:** Investors with large gaps between consecutive SIP transactions were identified as potentially at-risk, providing an opportunity for targeted investor engagement.
- **Portfolio Concentration:** HHI analysis showed differences in sector concentration among equity funds. Higher HHI values indicate greater exposure to a smaller number of sectors.

RECOMMENDATIONS

- Investors should consider both returns and risk when selecting mutual funds.
- Funds with high sector concentration should be monitored for concentration risk.
- Fund houses can use SIP continuation analysis to identify and engage at-risk investors.
- Risk-based fund recommendations can help investors identify suitable funds according to their risk appetite.
- Interactive dashboards should be used to continuously monitor fund performance, SIP trends and investor behaviour.

LIMITATIONS & CONCLUSION

LIMITATIONS

The analysis is based on the datasets provided for the project and therefore depends on their coverage, accuracy, and completeness. Historical performance and risk metrics do not guarantee future fund performance. Some investor and market factors may not be available in the datasets and therefore were not included in the analysis.

CONCLUSION

The Bluestock Mutual Fund Analytics project successfully combined Python, Pandas, SQLite, and Power BI to analyse mutual fund performance, risk, investor behaviour, SIP trends, and portfolio concentration.

The project produced advanced analytical outputs such as VaR, CVaR, rolling Sharpe ratio, investor cohorts, SIP continuation risk, fund recommendations, and sector HHI. The Power BI dashboard converted these findings into an interactive reporting solution with slicers, tooltips, and drill-through functionality.

Overall, the project demonstrates how data analytics can support better fund comparison, risk assessment, investor understanding, and data-driven decision-making.

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THANK YOU

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